



BIOMEDICAL IMPORTANCE OF CARBOHYDRATES IN HEALTH & DISEASE

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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Define carbohydrates and classify them
- Explain their biomedical importance
- Describe role in oral and systemic health
- Discuss carbohydrate-related diseases

DEFINITION OF CARBOHYDRATES

- Organic compounds containing C, H, and O
- General formula: $(CH_2O)_n$
- Major source of energy (4 kcal/g)
- Also called saccharides

CLASSIFICATION OF CARBOHYDRATES

1. Monosaccharides

- Glucose, Fructose, Galactose

2. Disaccharides

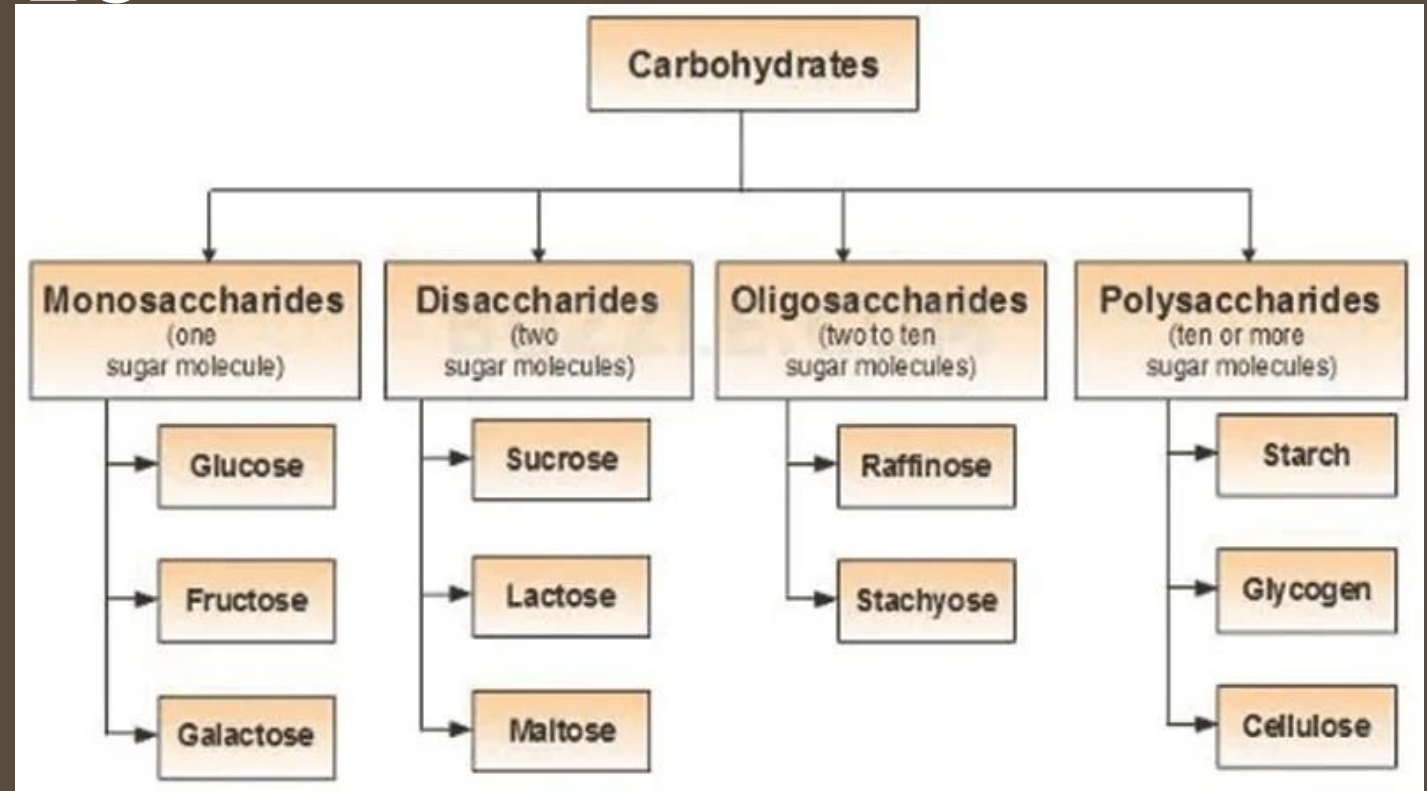
- Sucrose, Lactose, Maltose

3. Polysaccharides

* Starch

* Glycogen

* Cellulose



IMPORTANT MONOSACCHARIDES

- * Glucose - Blood sugar
- * Fructose - Fruit sugar
- * Galactose - Component of lactose

(Glucose is the most important biologically)

BIOMEDICAL IMPORTANCE - OVERVIEW

Carbohydrates are important for:

- * Energy production
- * Structural functions
- * Cell recognition
- * Metabolic pathways
- * Detoxification

CARBOHYDRATES AS ENERGY SOURCE

- * Primary fuel of the body
- * Brain depends mainly on glucose
- * RBCs depend entirely on glucose
- * Stored as glycogen in liver & muscle

STORAGE FORM - GLYCOGEN

- * Liver glycogen → maintains blood glucose
- * Muscle glycogen → energy during exercise
- * Glycogenolysis & Glycogenesis

STRUCTURAL ROLE

Carbohydrates form:

- * Glycoproteins
- * Glycolipids
- * Proteoglycans
- * Components of connective tissue

ROLE IN CELL RECOGNITION

- * Blood group substances (ABO antigens)
- * Cell membrane receptors
- * Immune response

ROLE IN NUCLEIC ACIDS

- * Ribose → RNA

- * Deoxyribose → DNA

DETOXIFICATION FUNCTION

- * Glucuronic acid conjugation
- * Bilirubin detoxification
- * Drug metabolism in liver

CARBOHYDRATES IN ORAL HEALTH

Dental Caries:

- * Caused by fermentation of dietary carbohydrates
- * Acid production by oral bacteria
- * Demineralization of enamel

Main organism: *Streptococcus mutans*

ROLE OF SUCROSE IN CARIES

- * Most cariogenic sugar
- * Forms sticky glucans
- * Enhances plaque formation

Dental Plaque Formation

- * Bacteria + salivary proteins + carbohydrates
- * Acid production lowers pH < 5.5
- * Enamel dissolution

CARBOHYDRATES IN SYSTEMIC DISEASES

DIABETES MELLITUS

- * Chronic hyperglycemia
- * Due to insulin deficiency/resistance
- * Disturbance in carbohydrate metabolism

Types of Diabetes

- * Type 1 - Insulin dependent
- * Type 2 - Insulin resistance
- * Gestational diabetes

ORAL MANIFESTATIONS OF DIABETES

- * Periodontitis
- * Delayed wound healing
- * Xerostomia
- * Increased caries risk

HYPOGLYCEMIA

- * Low blood glucose
- * Symptoms: sweating, tremors, confusion
- * May occur in insulin-treated patients

LACTOSE INTOLERANCE

- * Deficiency of lactase enzyme
- * GI symptoms after milk intake

Condition: Lactose intolerance

GALACTOSEMIA

- * Defect in galactose metabolism
- * Accumulation of galactose
- * Liver damage, cataracts

Disease: Galactosemia

GLYCOGEN STORAGE DISEASES (GSD)

- * Defect in glycogen metabolism
- * Excess or abnormal glycogen storage

Example: Von Gierke disease

IMPORTANCE IN NUTRITION

- * 55-60% of daily calories
- * Prefer complex carbohydrates(whole grains, legumes, vegetables, fruits)
- * Reduce refined sugars
- * Important for dental health counseling

RECOMMENDED DIETARY ADVICE FOR DENTAL PATIENTS

- * Limit frequency of sugar intake
- * Avoid sticky sweets
- * Rinse mouth after meals
- * Encourage fiber-rich diet

SUMMARY

- * Carbohydrates are major energy source
- * Essential for structural & metabolic functions
- * Important in oral and systemic health
- * Imbalance leads to diseases like diabetes & caries

